

Feature-engineered machine learning for daily-scale prediction of effluent total phosphorus and coagulant dosing optimization in full-scale DAF systems

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ABSTRACT

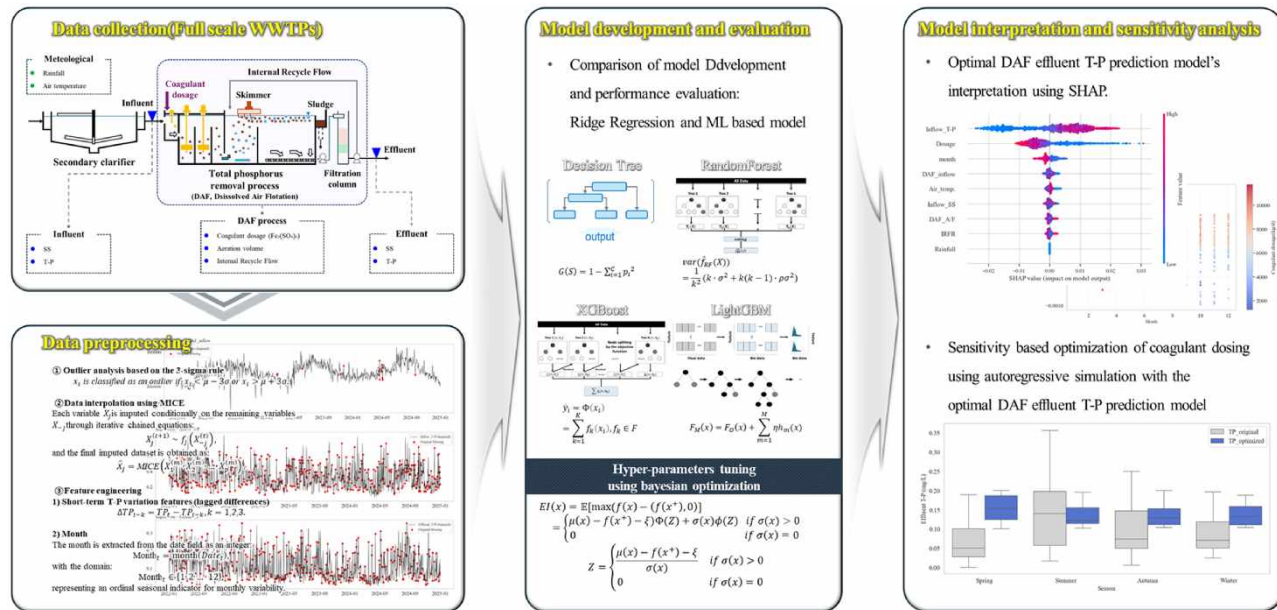
Effective phosphorus control in dissolved air flotation (DAF) systems is essential for regulatory compliance in full-scale wastewater treatment plants, yet operational decisions are often constrained to daily time scales due to limited online sensing. This study proposes an interpretable, feature-engineered machine-learning framework for daily prediction of effluent total phosphorus (T-P) and sensitivity-based optimization of coagulant dosing in a full-scale municipal DAF system (410,000 m³/day). Long-term operational, water-quality, and meteorological data (1,096 days) were preprocessed using three-sigma outlier screening and multivariate imputation by chained equations (MICE). Mechanistically informed features capturing influent loading, operational conditions, short-term variability of effluent T-P (1–3 day difference-based features), and seasonal effects were incorporated. Among the evaluated models, Random Forest achieved the best performance (Test $R^2 = 0.818$; RMSE = 0.032 mg/L), corresponding to a prediction error within 20% of the discharge limit (0.2 mg/L). SHAP analysis identified influent T-P, coagulant dosage, and short-term variation as dominant drivers across seasons. A sensitivity-based autoregressive simulation indicated that optimized dosing could reduce coagulant consumption by 32–51%, yielding an estimated annual cost saving of 1.53 billion KRW while improving effluent stability. The proposed framework demonstrates the practical value of daily-scale, interpretable machine learning for data-driven DAF operation.

Key words: coagulant optimization, DAF effluent total phosphorus prediction, data-driven tertiary treatment management, feature-engineered machine learning, SHAP-based model interpretability

HIGHLIGHTS

- Developed a feature-engineered ML framework for daily DAF effluent T-P prediction using long-term hydraulic, water-quality, and meteorological data.
- Random Forest showed the highest accuracy (Test $R^2 = 0.8175$).
- SHAP identified influent T-P, coagulant dosage, and short-term variation as key drivers.
- Sensitivity-based dosing optimization cut chemical use by 32–51% and stabilized effluent T-P.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Excessive phosphorus discharge from wastewater treatment plants (WWTPs) is widely recognized as a major contributor to eutrophication in receiving water bodies. To mitigate this environmental risk, South Korea incorporated total phosphorus (T-P) into national effluent standards in 1996 and subsequently strengthened region-specific discharge limits to 0.2–0.5 mg/L from 2012 onward (Kim *et al.* 1996; Seo *et al.* 2011; Kim & Chung 2022). These regulatory changes have substantially increased the operational pressure on municipal WWTPs, making stable compliance with effluent standards a fundamental operational requirement.

To meet tightened T-P discharge limits, physico-chemical phosphorus removal processes have been increasingly adopted to complement biological nutrient removal, which alone is often insufficient under stringent regulatory conditions (Hu *et al.* 2012; Bunce *et al.* 2018; Abdoli *et al.* 2024; Alatabe 2024). Among these technologies, dissolved air flotation (DAF) has been widely applied as an advanced or tertiary treatment due to its high clarification efficiency, compact footprint, and suitability for retrofitting in space-limited facilities (Edzwald 2010; Kwak & Lee 2015; Lee *et al.* 2016). Both pilot- and full-scale studies have demonstrated that, when appropriate coagulation and flotation conditions are maintained, DAF can achieve consistently low effluent T-P concentrations, confirming its effectiveness for phosphorus control in municipal WWTPs (Maeng *et al.* 2017; Väänänen *et al.* 2023). In addition, pilot-scale studies integrating DAF with complementary physico-chemical or biological processes, such as algal photo-bioreactors, have demonstrated effective removal of chemical oxygen demand (COD), nutrients, and suspended solids, highlighting the flexibility of DAF as part of hybrid advanced treatment systems (Badawi *et al.* 2022).

Despite its proven performance, DAF operation is highly sensitive to influent water quality variability and coagulant dosing conditions. Insufficient coagulant dosing frequently results in effluent T-P exceedances, whereas excessive dosing leads to unnecessary chemical consumption and increased sludge production (Di Bella *et al.* 2014; Khettaf *et al.* 2021; Narges *et al.* 2021; Wang *et al.* 2021a). Consequently, determining an optimal coagulant dosage represents a critical operational challenge from both economic and regulatory perspectives. Nevertheless, many WWTPs still rely on empirical strategies such as fixed-ratio or flow-proportional dosing, which often deviate from optimal process conditions when influent phosphorus concentration, turbidity, and hydraulic loading fluctuate rapidly (Álvarez Díez *et al.* 2024).

To address these limitations, recent studies have explored data-driven and intelligent approaches for coagulation process optimization (Liu *et al.* 2024; Lu *et al.* 2024; Sheik *et al.* 2024). In particular, machine-learning models have been successfully applied to large-scale, long-term operational datasets from full-scale WWTPs to predict effluent total phosphorus,

demonstrating the feasibility of data-driven water-quality forecasting under highly variable influent and process conditions (Ly *et al.* 2022). Machine-learning-based models have been proposed to predict optimal coagulant dosage using historical and operational data, demonstrating improved robustness and efficiency compared with conventional empirical control strategies (Wang *et al.* 2021b; Fang *et al.* 2022; Tochio *et al.* 2023; Kim *et al.* 2024a, b).

Despite these advances, the practical deployment of real-time control strategies operating at minute- or hourly-scale remains limited in full-scale municipal WWTPs. This is mainly due to sensor reliability issues, maintenance demands, and the difficulty of integrating advanced control systems with existing operational infrastructures (Pilz *et al.* 2024; Li *et al.* 2025). As a result, most municipal WWTPs are not operated under continuous real-time control but rather rely on daily laboratory-based effluent measurements and routine operational records for process management (Alsulaili & Refaie 2021; Afan *et al.* 2024). Under such realistic operating conditions, operational decisions related to chemical dosing are typically made at a daily or shift-based time scale.

From this perspective, daily-scale prediction models represent a pragmatic and operationally relevant alternative to minute- or hourly-scale real-time optimization strategies, particularly in facilities where online phosphorus sensors or fully automated dosing systems are not yet implemented (Guo *et al.* 2020; Cechinel *et al.* 2024; Lai 2025). However, many existing daily-scale prediction studies rely on black-box models with limited interpretability, which restricts their applicability to practical decision-making in WWTP operation.

The primary objective of this study is to develop an interpretable machine-learning framework for daily prediction of DAF effluent T-P concentrations in full-scale municipal wastewater treatment plants. The proposed framework integrates water-quality factors, DAF operational factors, and meteorological factors with engineered features that capture short-term phosphorus variability and seasonal patterns, thereby reflecting process dynamics under real operating conditions. Rather than focusing solely on predictive accuracy, this study emphasizes model interpretability to identify key operational drivers and to support informed decision-making in coagulant dosing management. In addition, the predictive framework is used to examine the sensitivity of effluent T-P concentrations to changes in coagulant dosage, illustrating its potential to reduce chemical consumption while maintaining stable regulatory compliance. By combining daily-scale prediction, interpretability, and operational relevance, this study aims to advance the practical application of machine learning in physicochemical wastewater treatment and sustainable utility management.

2. METHODS

This study developed a daily-scale prediction and scenario-based optimization framework for effluent T-P in a full-scale DAF process. The overall methodological workflow is organized into three sequential stages, as illustrated in Figure 1.

The first stage focuses on data preprocessing, including multi-source data acquisition, outlier screening, missing-value imputation, and feature engineering to represent process mechanisms, short-term variability, and seasonal effects in daily DAF operation. The second stage involves model development and evaluation, in which machine-learning models are trained using the engineered feature set, optimized through Bayesian hyperparameter tuning, and assessed using standard regression performance metrics. The third stage integrates model interpretation and operational optimization, where SHapley Additive exPlanations (SHAP) are applied to quantify feature contributions and interactions, and a scenario-based autoregressive simulation framework is used to evaluate the sensitivity of effluent T-P to coagulant dosing. Together, these stages provide an interpretable and actionable decision-support framework for full-scale DAF operation.

2.1. Data acquisition

The target municipal WWTP has a design capacity of 410,000 m³/day and operates with two parallel treatment lines. The main biological treatment process follows the Modified Ludzack–Ettinger (MLE) configuration, after which the secondary effluent is conveyed to tertiary treatment units comprising a micro disk filter (MDF) and a DAF system operated in parallel.

Among these tertiary treatment processes, the present study focuses on the full-scale DAF system, which serves as the primary physicochemical unit for T-P removal. DAF is a physicochemical solid–liquid separation process in which air is dissolved into water under high pressure and subsequently released to generate fine microbubbles through rapid depressurization. In accordance with Henry's law, increased pressure enhances air solubility, allowing the treated water to become air-saturated in the pressurization unit. The generated microbubbles attach to chemically destabilized colloidal particles and coagulated flocs formed by ferric sulfate dosing, imparting buoyancy to the aggregates. These bubble–floc complexes rise

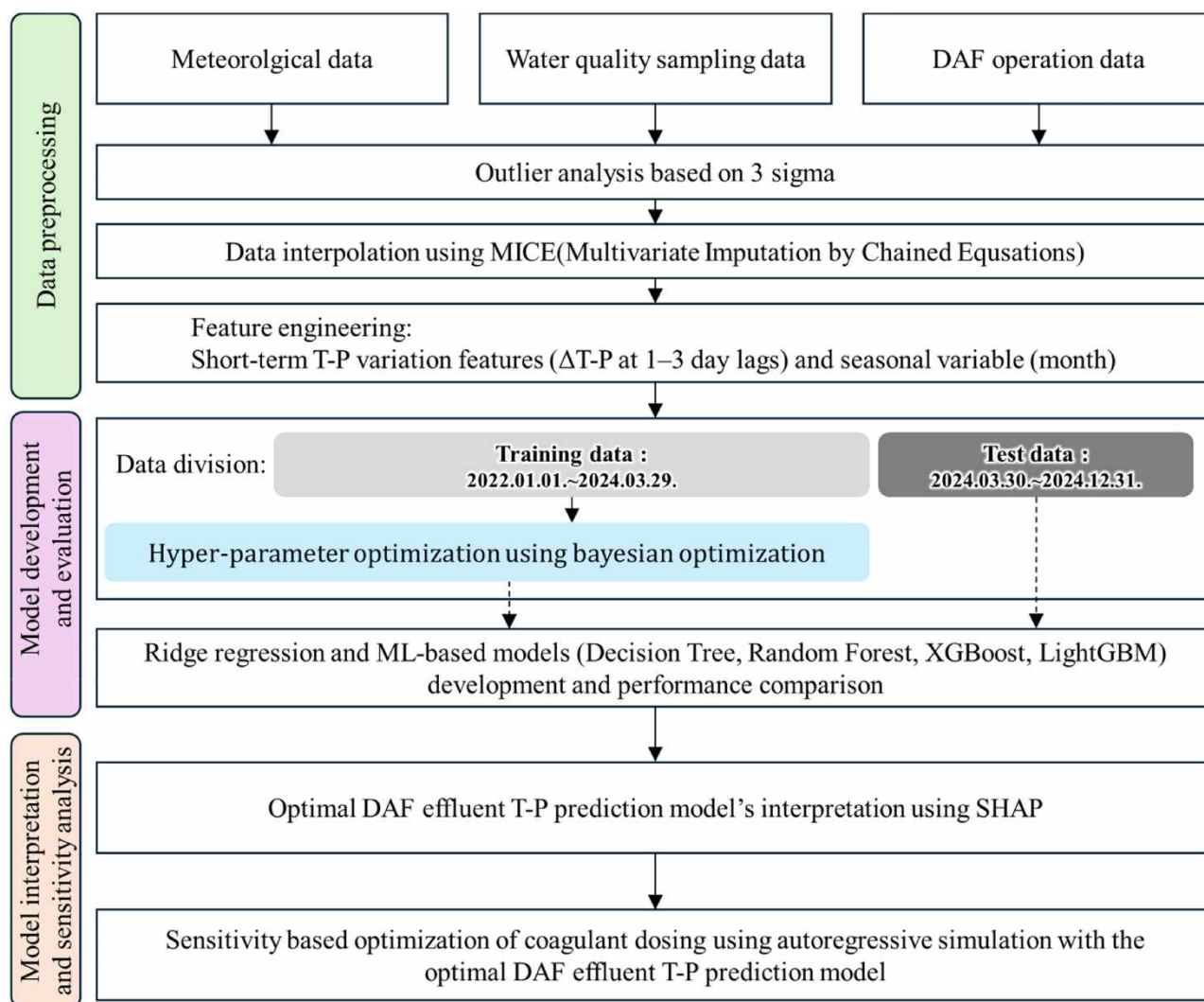


Figure 1 | Schematic diagram of research.

to the water surface and are removed by surface skimming, thereby achieving effective solid–liquid separation and phosphorus removal.

The DAF process is continuously operated at full scale and employs ferric sulfate ($\text{Fe}_2(\text{SO}_4)_3$) as the coagulant, with daily operational decisions typically based on laboratory-scale water-quality analyses rather than real-time sensor feedback. Table 1 provides a consolidated overview of the key design, operational, and monitoring characteristics of the target WWTP and the associated DAF system, thereby summarizing the study context in terms of treatment configuration, operational scale, control strategy, and data coverage.

The dataset used in this study was compiled from three independent data sources collected at the target WWTP over the period from January 2022 to December 2024 ($N = 1,096$ days). Meteorological data were obtained from the Automated Synoptic Observing System (ASOS) operated by the Korea Meteorological Administration. Water-quality data, including influent and effluent T-P, suspended solids (SS), pH, and internal recycle flow rate (IRFR), were obtained through routine daily laboratory analyses. In addition, DAF operational variables – such as influent flow rate, coagulant dosage, mixing conditions, recycle ratio, and air-to-solids ratio – were extracted from the plant's Supervisory Control and Data Acquisition (SCADA) system.

Figure 2 presents a schematic overview of the DAF process and the corresponding data collection points, illustrating the coagulation–flocculation–flotation sequence and the interactions among influent characteristics, operational conditions, and effluent quality responses under full-scale operating conditions.

Table 1 | Key design and operational characteristics of the full-scale DAF system used as the tertiary phosphorus removal unit in the target WWTP

Category	Description
Location	Municipal WWTP in South Korea
Design capacity	410,000 m ³ /day
Biological process	Modified Ludzack–Ettinger (MLE)
Tertiary treatment	DAF (parallel operation)
Role of DAF	Primary physicochemical phosphorus removal
Coagulant type	Ferric sulfate (Fe ₂ (SO ₄) ₃)
Operation scale	Full-scale, continuously operated
Control strategy	Daily operational decision based on laboratory analysis
Data period	2022.01.01.–2024.12.31. (N = 1,096 days)

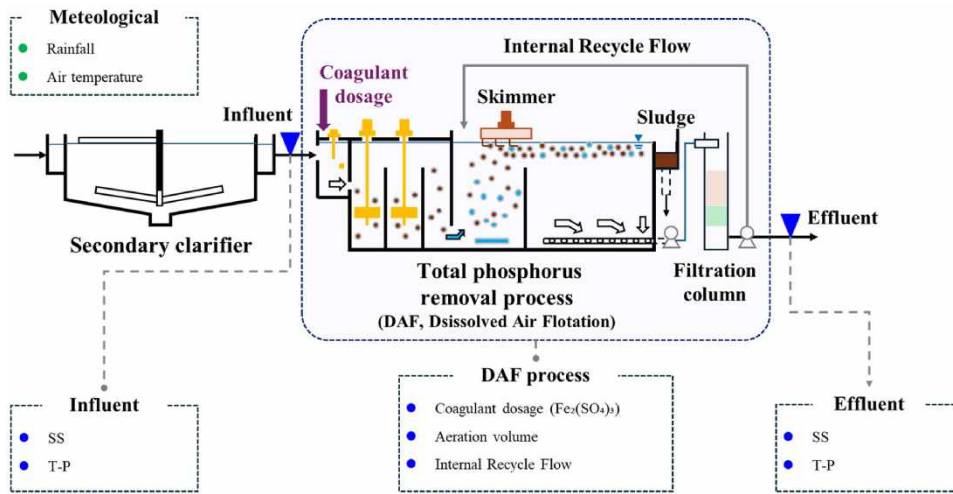


Figure 2 | Schematic diagram of the DAF process and the corresponding data collection.

2.2. Data preprocessing

Prior to model development, outlier detection was conducted using a three-sigma rule applied to all major operational and water-quality parameters. Records exceeding $\pm 3\sigma$ from the mean were considered abnormal events caused by sensor errors, equipment malfunction, or extreme operational disturbances, and were removed to ensure stable model training.

Missing values were imputed using the Multivariate Imputation by Chained Equations (MICE) algorithm, in which each variable with missing entries X_j is modeled conditionally on all other variables X_{-j} using a regression-based conditional distribution:

$$X_j^{\text{mis}} \sim f_j(X_{-j}; \theta_j).$$

These conditional models are iteratively updated following a chained-equation scheme:

$$\hat{X}^{(t)} = \mathcal{F}(\hat{X}^{(t-1)}),$$

where $\hat{X}^{(t)}$ denotes the imputed dataset at iteration t .

This iterative process continues until convergence, producing a completed dataset that preserves the multivariate correlation structure among meteorological, operational, and water-quality variables.

To develop a daily-scale prediction model capable of capturing the complex behaviour of effluent T-P in the DAF process, an extensive feature-engineering strategy was implemented to incorporate both mechanistic process knowledge and the statistical characteristics of real plant operation. In full-scale systems, effluent T-P exhibits pronounced variability arising from multiple interacting factors, including fluctuations in coagulant dosing here, ferric sulfate ($\text{Fe}_2(\text{SO}_4)_3$) used as the primary coagulant rainfall-driven changes in influent dilution and particulate loading, seasonal variations in temperature that modify coagulation and flotation reaction kinetics, operational conditions related to mixing intensity, micro-bubble availability, and solids loading, as well as event-driven disturbances in influent quality.

To represent these nonlinear and multiscale dynamics more effectively than with simple lag inputs alone, the feature-engineering process incorporated short-term variation metrics based on one-, two-, and three-day differences in effluent T-P. These difference-based features capture the direction and magnitude of rapid fluctuations that commonly arise from disturbance-driven conditions such as rainfall-induced influent changes, temporary under-dosing of coagulants, or shifts in particulate characteristics. By summarizing immediate transitions and short-horizon trends, these descriptors allow the model to better detect localized instability and deviation from typical operating patterns.

Importantly, all short-term variation features were computed exclusively using information available up to the prediction timestamp, ensuring that no future observations were used in feature construction and thereby avoiding information leakage in the daily operational setting. As illustrated in the feature-engineering schematic, the final input set therefore consisted of short-term T-P variation features ($\Delta\text{T-P}$ at 1–3 day lags), which capture rapid disturbance-driven dynamics, together with a seasonal indicator (month) to represent longer-term cyclic effects in DAF operation. This design is consistent with variable delay-time and sensitivity analyses reported in related water and groundwater modelling studies, where explicitly screening lagged and differenced inputs has been shown to improve predictive robustness while preserving operational realism (Sharghi *et al.* 2025). As a result, they enhance the model's sensitivity to abrupt changes and improve its ability to distinguish stable periods from disturbance-prone intervals in daily DAF operation.

In addition, process-mechanistic operational indicators including the air-to-inflow ratio (DAF_{A/F}), ferric-sulfate dosing rate, influent flow, and the IRFR were incorporated to reflect structural relationships among operational conditions, bubble saturation, mixing performance, and phosphorus separation efficiency.

External environmental drivers such as air temperature, rainfall intensity, and a monthly seasonal index were also included to capture climate-dependent variations in hydraulic loading and coagulation–flotation behaviour. Finally, raw-water quality indicators such as influent T-P and influent SS were integrated to account for upstream water-quality conditions that directly influence coagulant demand and flotation efficiency. Collectively, this feature-engineering framework provides a unified representation of process mechanisms, short-term dynamics, seasonality, and disturbance signals, effectively addressing the inherent non-stationarity and high variability of daily DAF operation and thereby improving the robustness, predictive performance, and interpretability of the machine-learning models developed in this study.

2.3. Development of ML-based models for DAF effluent T-P prediction model

To construct a reliable daily-scale prediction model for DAF effluent T-P, four representative machine-learning algorithms Decision Tree, Random Forest, XGBoost and LightGBM were developed and evaluated. These models were selected to compare different tree-based learning paradigms, including a single interpretable tree structure (Decision Tree), a bootstrap aggregation ensemble (Random Forest), a gradient-boosting framework with regularization (XGBoost), and a histogram-based gradient-boosting algorithm optimized for large-scale tabular environmental data (LightGBM).

Hyperparameter tuning was performed using a Bayesian optimization framework to efficiently explore the parameter space while minimizing computational cost. The search space for each model followed the ranges summarized in Table 2. For all models, the objective was to minimize the mean squared error (MSE) between observed and predicted effluent T-P concentrations.

2.4. Model's performance evaluation

To evaluate the predictive accuracy of the developed machine-learning models, three widely used regression metrics R^2 , RMSE, and MAE were employed. R^2 measures the proportion of variance in observed T-P concentrations that is explained by the model, with higher values indicating stronger explanatory power. RMSE quantifies the overall magnitude of prediction errors and is particularly sensitive to large deviations, thereby reflecting the model's ability to avoid substantial under- or over-prediction. MAE represents the average absolute difference between observed and predicted values and provides a

Table 2 | Hyper-parameter range of ML models

Division	Decision Tree	Random Forest	XGBoost	Light GBM
Max depth		2 ~ 10 (Integer)		
Min samples split	2 ~ 20 (Integer)		–	
Min samples leaf	1 ~ 50 (Integer)		–	
N_estimators		100 ~ 2,000 (Integer)		
Learning rate	–	–	0.001 ~ 0.1 (float)	
Min child weight	–	–	5.0 ~ 15.0 (float)	–
Min child samples	–	–		50 ~ 250 (Integer)
Gamma	–	–	0.0 ~ 3.0 (float)	–
Subsample	–	–	0.6 ~ 1.0 (float)	
Colsample_bytree	–	–	0.6 ~ 1.0 (float)	
Number leaves	–	–		5 ~ 30 (Integer)
Loss function	MSE minimize			

more robust measure against outliers compared with RMSE. Collectively, these metrics provide a comprehensive assessment of both the accuracy and stability of the effluent T-P prediction models.

The evaluation metrics were computed as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

where,

y_i = observed effluent T-P concentration,

$\hat{y}_i$ = model-predicted effluent T-P concentration,

$\bar{y}$ = mean of observed values,

n = number of samples.

2.5. Feature contribution and interaction analysis using SHAP

To interpret the trained DAF effluent T-P prediction model, SHapley Additive exPlanations (SHAP) was applied. SHAP quantifies the contribution of each input feature to a model prediction by computing Shapley values, which represent the marginal effect of adding a feature to all possible feature subsets. The Shapley value for feature i is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

where,

F = full set of features,

S = any subset of F excluding feature i ,

$f(\cdot)$ = model prediction for a given feature set.

For the optimal model, two SHAP outputs were analysed: (1) Global feature contribution, based on mean absolute Shapley value, to identify key drivers affecting effluent T-P; (2) Feature interaction patterns, using pairwise SHAP interaction values, to examine nonlinear dependencies between major operational factors. This SHAP-based analysis enabled a transparent interpretation of how individual features and their interactions influence T-P predictions, providing mechanistic insight into the DAF process.

2.6. Sensitivity-based optimization of coagulant dosing

In this study, a sensitivity-based optimization framework was implemented to evaluate how changes in coagulant dosage influence effluent T-P concentrations and to identify the minimum dosing level required to satisfy the target effluent standard. Using the machine-learning-based DAF effluent T-P prediction model, dynamic simulations were conducted to quantify the system's response to incremental adjustments in coagulant dosing intensity.

To simulate the effect of dosing modifications, a multiplier m was applied to the baseline coagulant dosage at each time step t , generating a series of hypothetical operational scenarios. The autoregressive structure was then employed to propagate the impact of dosage adjustments over time. Specifically, short-term variation features were recalculated at each prediction step using the model's preceding predicted values rather than observed measurements. This recursive updating scheme enables the simulation to capture cumulative process responses and nonlinear feedback effects that arise when dosing levels deviate from the baseline operational condition.

Through this autoregressive, multiplier-based simulation approach, the sensitivity of effluent T-P to different dosing intensities was quantified, enabling the determination of optimal and suboptimal dosing ranges that balance regulatory compliance and chemical usage.

3. RESULTS AND DISCUSSION

3.1. Statistical analysis of characteristics of the target DAF system

Figure 3 presents the violin plots and histograms of major hydraulic and water-quality parameters including DAF inflow, influent pH, influent SS, influent T-P, and effluent T-P before and after outlier removal based on the three-sigma rule. Overall, the cleaned distributions exhibit reduced extreme values and smoother probability densities, indicating that abnormal spikes caused by operational noise, laboratory errors, or temporary disturbances were effectively removed. Importantly, the overall shapes of the histograms demonstrate that the outlier filtering procedure did not distort the original data structure; instead, it selectively removed anomalous points while preserving the inherent distributional characteristics of the dataset.

Consistent with this, the DAF inflow distribution shows only minimal changes after cleaning, reflecting the inherent stability of hydraulic loading data and confirming that outlier removal did not alter the natural variability of flow patterns. In contrast, more noticeable improvements were observed in influent SS, influent T-P, and effluent T-P, where long-tailed and skewed distributions became more compact and symmetric after removing extreme observations. These refinements indicate that the applied filtering effectively removed unrepresentative spikes without modifying the central tendency or typical operating ranges.

Taken together, the results demonstrate that the three-sigma-based outlier removal method successfully eliminated irregular measurement spikes while maintaining the essential statistical characteristics of the original dataset, ensuring reliable and unbiased input for downstream machine-learning modelling.

Table 3 summarizes the extent of missing data and the number of observations removed as outliers for each candidate variable. Overall, the proportion of outliers identified by the three-sigma rule was low across all variables, indicating that extreme values were rare and limited to a small number of observations. Missing data was mainly concentrated in influent pH, reflecting intermittent sensor downtime, whereas most hydraulic, operational, and meteorological variables exhibited negligible missingness. These results indicate that the applied outlier removal procedure was limited in scope and did not substantially alter the overall data distribution.

Table 4 summarizes the descriptive statistics of the candidate input variables after outlier removal using the three-sigma rule. Overall, the cleaned dataset exhibits stable central tendencies and realistic operational ranges, confirming that the remaining values are representative of typical DAF process behaviour and are suitable for use as input features in model development.

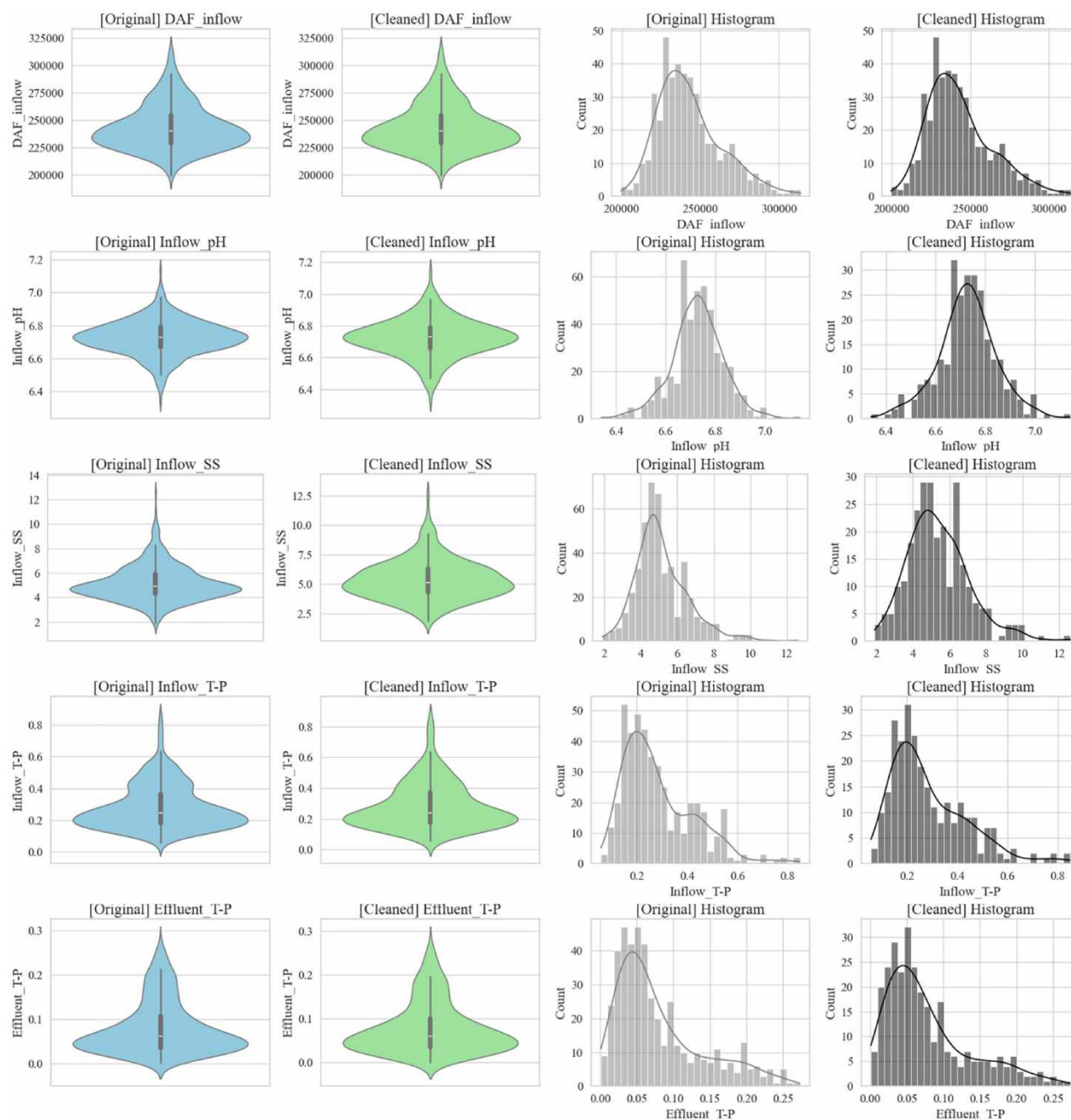


Figure 3 | Violin plots and histograms of influent flow rate and key water-quality features after 3-sigma-based outlier removal.

For the hydraulic loading group, the DAF inflow shows a mean of $240,671 \text{ m}^3/\text{d}$ with moderate variability ($\text{std} = 23,394 \text{ m}^3/\text{d}$), reflecting regular fluctuations in plant-scale daily flow while excluding extreme peaks removed during the pre-processing stage. The IRFR exhibits a mean of 36.09% and a narrow interquartile range, consistent with its function of maintaining dissolved-air saturation levels within operational norms.

Among the influent water-quality variables, influent SS and influent T-P retained sufficient variability and complete records to serve as meaningful predictors for downstream phosphorus behaviour. In contrast, influent pH, although stable within a narrow range (approximately 6.7–7.1), exhibited more than 30% missing observations due to intermittent sensor downtime. Given its low temporal variability and high missing-value rate, influent pH was excluded from the candidate feature set used for model development.

Table 3 | Missing data and outlier statistics after three-sigma filtering

Group	Factors	Unit	Missing data		Outlier data	
			Count	Rate	Count	Rate
Flow	DAF Inflow	m ³ /d	4	0.36%	2	0.18%
	IRFR ^a	%	6	0.55%	4	0.36%
Water quality	Inflow pH	–	479	43.70%	1	0.09%
	Inflow SS	mg/L	68	6.20%	6	0.55%
	Inflow T-P	mg/L	59	5.38%	6	0.55%
	Effluent T-P	mg/L	74	6.75%	9	0.82%
Operation	DAF A/F ^b	–	4	0.36%	4	0.36%
	Coagulant dosage ^c	kg/d	0	0.00%	1	0.09%
Meteorological	Rainfall	mm	0	0.00%	0	0.00%
	Air temp.	°C	0	0.00%	0	0.00%

^aDAF Internal Recycle Flow Rate: A process parameter that returns a portion of the treated effluent to the DAF saturator to increase dissolved air concentration and improve flotation performance.

^bDAF A/F: An operational indicator expressing the ratio of dissolved air to influent flow, where bubble diameter and flotation velocity are proportional to the amount of dissolved air; commonly used to assess the saturation ratio in DAF systems.

^cCoagulant dosage: A key operational factor in tertiary phosphorus removal processes, representing the dosing relative to influent hydraulic and pollutant loading, which directly influences the efficiency of total phosphorus precipitation and removal.

Table 4 | Statistical description of features after outlier removal

Group	Factors	Unit	mean	std	min	25%	50%	75%	max
Flow	DAF Inflow	m ³ /d	240,671	23,394	131,176	225,232	240,000	256,992	313,936
	IRFR	%	36.09	4.86	25.14	32.52	35.63	38.83	62.14
Water quality	Inflow pH	–	6.73	0.12	6.34	6.66	6.73	6.79	7.14
	Inflow SS	mg/L	5.26	1.74	1.00	4.10	5.00	6.30	11.10
	Inflow T-P	mg/L	0.32	0.15	0.06	0.20	0.28	0.42	0.87
	Effluent T-P	mg/L	0.10	0.07	0.00	0.05	0.08	0.14	0.36
Operation	DAF A/F	–	0.01	0.00	0.01	0.01	0.01	0.01	0.02
	Coagulant dosage	kg/d	6,267	1,899	720	4,760	5,980	7,680	11,750
Meteorological	Rainfall	mm	1.44	7.42	0.00	0.00	0.00	0.00	122.50
	Air temp.	°C	14.09	10.70	– 14.70	4.80	15.70	23.50	31.80

In effluent quality, the stability of the DAF process is reflected in the relatively narrow variability of effluent T-P, which shows a mean concentration of 0.10 mg/L with most values clustered below 0.14 mg/L. The maximum value of 0.36 mg/L suggests that episodic performance variations remain captured without retaining unrealistic outlying values.

For operation-related indicators, the DAF A/F ratio remains around 0.01 across all quartiles, aligning with typical saturation conditions observed in full-scale dissolved air flotation. Coagulant dosage also shows a stable and well-bounded distribution (mean = 6,267 kg/d, IQR = 4,760–7,680 kg/d), indicating that the dosing data are sufficiently reliable to be used as an operational control variable in the machine-learning model.

Lastly, meteorological variables exhibit naturally large variability, particularly rainfall (0–122.5 mm/d), which is important for modelling storm-induced influent dilution and loading changes. Air temperature ranges from –14.7 to 31.8 °C, capturing seasonal dynamics that influence coagulation and flotation behaviour.

Collectively, the variables summarized in Table 1 demonstrate well-structured distributions after outlier filtering, ensuring that the selected features retain operational meaning while eliminating noise. These characteristics support their suitability as candidate inputs for developing robust daily-scale DAF effluent T-P prediction models.

3.2. MICE-based interpolation of input and target features

Following outlier removal, missing values in both input and target variables were imputed using the MICE algorithm to preserve the multivariate relationships among hydraulic, water-quality, operational, and meteorological parameters. MICE

sequentially estimate missing values through iterative regression models, allowing each variable with missing entries to be predicted from all other available variables. This approach is particularly suitable for environmental process data, where interactions between influent loading, operational conditions, and effluent responses exhibit nonlinear and interdependent patterns.

Figure 4 illustrates the imputation results for key variables, including DAF inflow, influent SS, influent T-P, and effluent T-P.

The red markers represent originally missing observations, while the continuous black lines indicate the reconstructed values obtained through MICE. Across all variables, imputed values align smoothly with surrounding temporal trends, without introducing abrupt discontinuities or irregular fluctuations, indicating that the underlying temporal structure was reasonably preserved. For instance, missing points in influent SS and influent T-P, which frequently occur during peak loading periods or laboratory analysis gaps, were reconstructed in a manner consistent with the observed variability of adjacent data. Similarly, missing effluent T-P values were imputed along realistic daily and seasonal fluctuations, ensuring continuity in the target variable for model training.

In addition, the use of multivariate imputation approaches such as MICE is supported by previous studies on environmental and water-quality datasets, which have shown that multivariate methods generally outperform simple interpolation-based techniques when handling mixed-type and process-driven missing data (Dixneuf *et al.* 2021). Similar findings have also been reported in water-quality monitoring studies, indicating that multivariate approaches, including MICE and tree-based methods, are more suitable for reconstructing longer-term or structurally missing segments than simple interpolation methods (Wang *et al.* 2024). Overall, the MICE-based imputation procedure maintained the general statistical characteristics and temporal coherence of the dataset, providing a complete and consistent input dataset for subsequent machine-learning model development.

3.3. Development of the optimal DAF effluent T-P prediction model

Figure 5 presents the Pearson correlation coefficients between effluent T-P and the candidate input features considered for model development. Overall, the correlations reveal meaningful linear associations that reflect the mechanistic behaviour of the DAF system.

Among the variables, influent T-P exhibited the strongest positive correlation with effluent T-P, indicating that higher upstream phosphorus loading tends to increase the residual phosphorus concentration after flotation. Influent SS also showed a moderate positive correlation, consistent with the role of particulate matter as a carrier of phosphorus species that affects coagulation and flotation performance.

In contrast, coagulant dosage demonstrated a clear negative correlation with effluent T-P. This relationship reflects the expected chemical-removal mechanism wherein higher doses of ferric coagulant enhance phosphorus precipitation and reduce the remaining phosphorus in the effluent. The strength of this negative correlation underscores the operational importance of appropriate dosing control.

Hydraulic indicators such as DAF inflow and IRFR showed only weak correlations with the target variable, implying that flow fluctuations or dissolved-air recirculation ratios have limited direct linear influence on effluent T-P under normal operational conditions. Similarly, DAF_A/F exhibited little linear association, which is reasonable given its narrow operational range in full-scale practice.

Meteorological variables including air temperature and rainfall showed weak but directionally consistent correlations. Air temperature displayed a slightly positive association, possibly reflecting seasonal variations in reaction kinetics or influent loading patterns. Rainfall exhibited a weak positive correlation as well, aligning with dilution and wash-off effects during wet-weather events.

The month variable showed only minimal correlation with the target, indicating that seasonality alone does not strongly determine effluent T-P levels unless coupled with other factors such as dosing or influent variability.

Table 5 shows that nonlinear machine-learning models clearly outperform the linear ridge regression model in predicting daily effluent T-P concentrations. Ridge regression exhibited limited explanatory power (Test $R^2 = 0.5599$), demonstrating that linear approaches are not suitable for wastewater treatment processes characterized by strong nonlinearities and complex interactions among hydraulic, operational, and water-quality variables. In contrast, tree-based ensemble models substantially improved predictive accuracy by effectively capturing these nonlinear behaviours. Among them, Random Forest achieved the best overall performance, recording the highest test R^2 (0.8175) and the lowest RMSE (0.0324 mg/L).

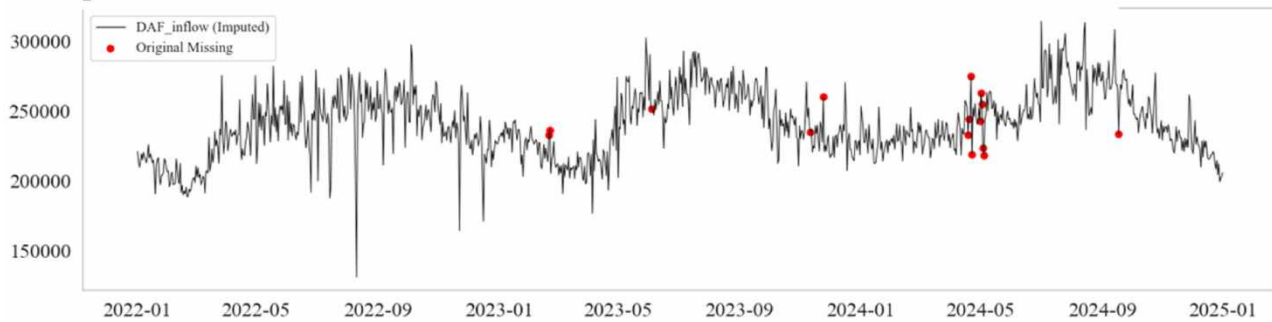
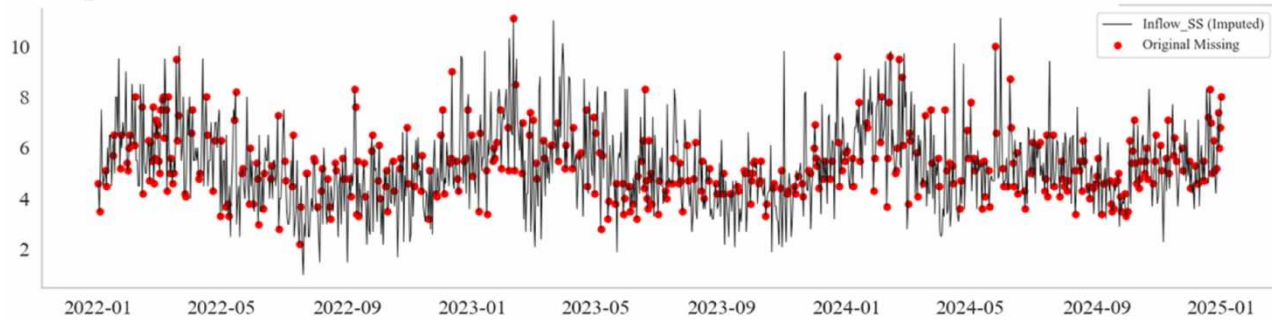
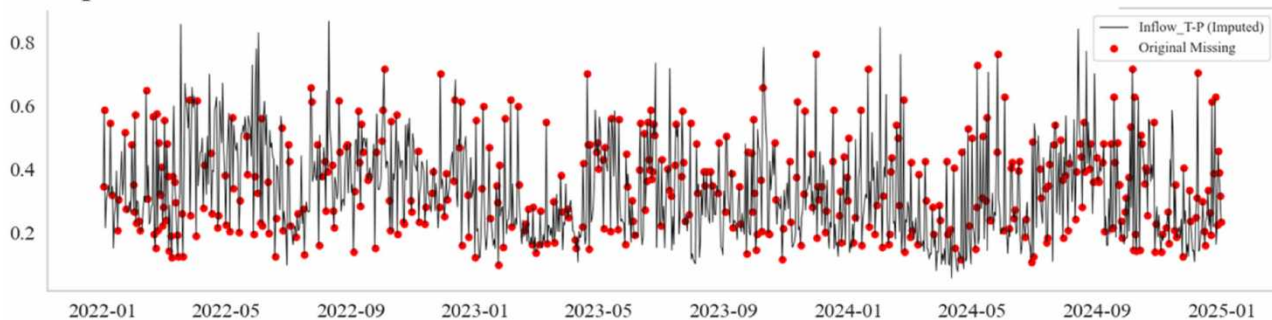
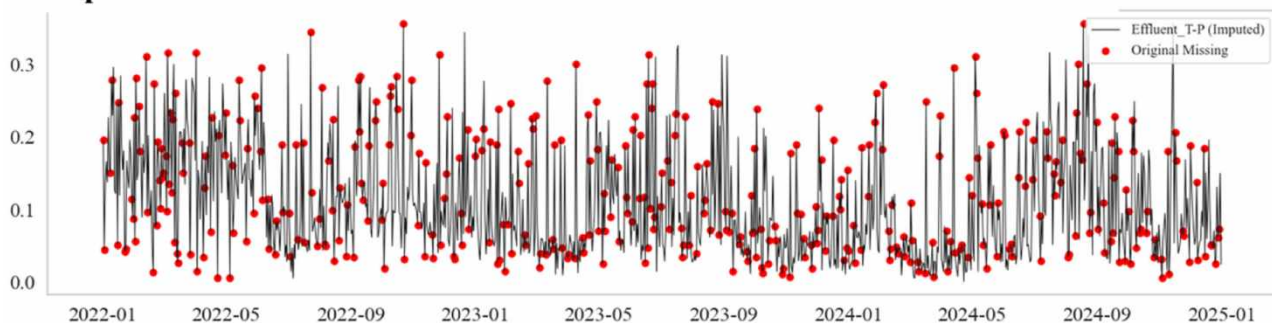
(a) Imputation results for DAF inflow**(b) Imputation results for DAF inflow SS concentration****(c) Imputation results for DAF inflow T-P concentration****(d) Imputation results for DAF effluent T-P concentration**

Figure 4 | MICE-based missing data interpolation results for influent flow and key water-quality variables.

and MAE (0.0235 mg/L). This superior generalization ability reflects the strengths of bootstrap aggregation and random feature sampling, which reduce overfitting and enhance model robustness.

From a regulatory and operational perspective, the achieved prediction accuracy should be interpreted relative to the effluent T-P discharge standard of 0.2 mg/L applied to the study site. Under this constraint, the Random Forest model's RMSE (0.0324 mg/L) and MAE (0.0235 mg/L) correspond to approximately 16 and 12% of the discharge limit, respectively,

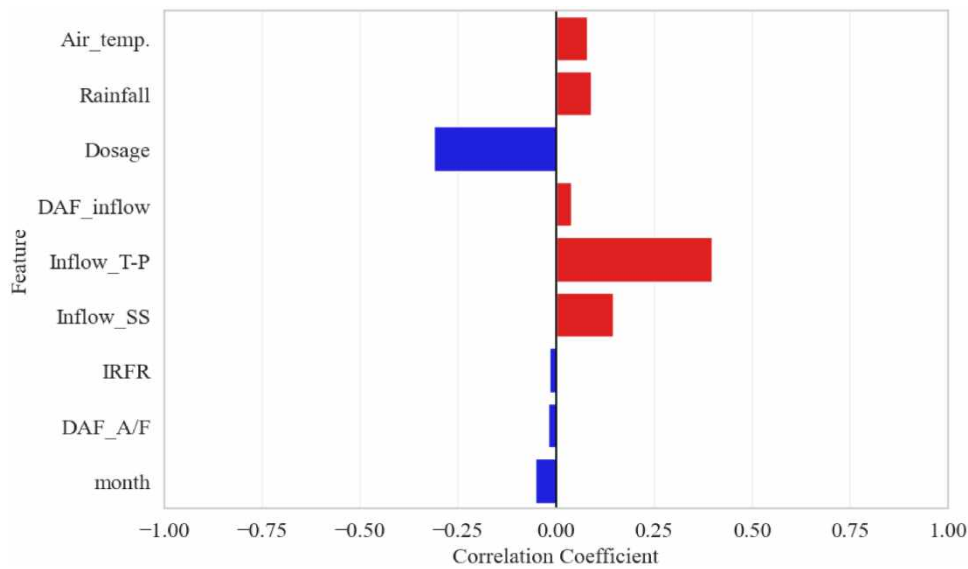


Figure 5 | Bar plots of Pearson correlation coefficients illustrating the linear associations between each input feature and the target variable.

Table 5 | Comparative performance evaluation of Ridge regression and ML models

Division	Ridge regression		Decision Tree		Random Forest		XGBoost		Light GBM	
	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
R ²	0.6706	0.5599	0.9048	0.7578	0.8678	0.8175	0.8339	0.7924	0.8374	0.7960
RMSE	0.0436	0.0504	0.0234	0.0374	0.0276	0.0324	0.0310	0.0346	0.0306	0.0343
MAE	0.0352	0.0380	0.0172	0.0276	0.0209	0.0235	0.0238	0.0259	0.0232	0.0256

implying that more than 80% of the allowable concentration range is resolved without prediction error. In practical terms, this indicates that the model achieves an effective accuracy exceeding 80% when evaluated against the regulatory criterion, providing sufficient resolution to support compliance-oriented operation and timely identification of potential exceedance risks.

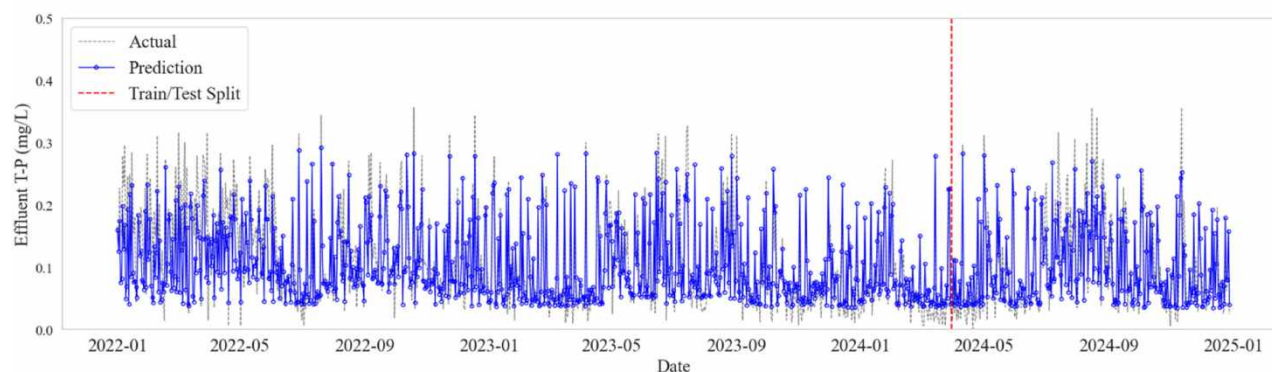
Although the test R² values of LightGBM (0.7960) and XGBoost (0.7924) are numerically close to that of Random Forest, the consistently lower RMSE and MAE achieved by Random Forest reduce absolute uncertainty near the regulatory threshold. When effluent T-P concentrations approach 0.2 mg/L, even small differences on the order of 0.01–0.02 mg/L may influence whether preventive coagulant dosing is initiated or delayed. Therefore, the observed performance advantage, while modest in aggregate metrics, becomes practically significant for compliance-oriented decision-making and conservative chemical dosing strategies, where minimizing the risk of underestimating peak concentrations is critical.

The exceptional performance of Random Forest is consistent with previous findings in the literature. [Sun *et al.* \(2024\)](#) reported that Random Forest Regression achieved very high R² values across multiple effluent parameters and was among the most reliable models for WWTP prediction. Additional studies applying ensemble bagging methods for WWTP performance evaluation have also shown that bagging-based models perform as well as, or better than other commonly used machine-learning models, owing to their ability to reduce variance and effectively capture the complex, nonlinear structures inherent in wastewater treatment processes.

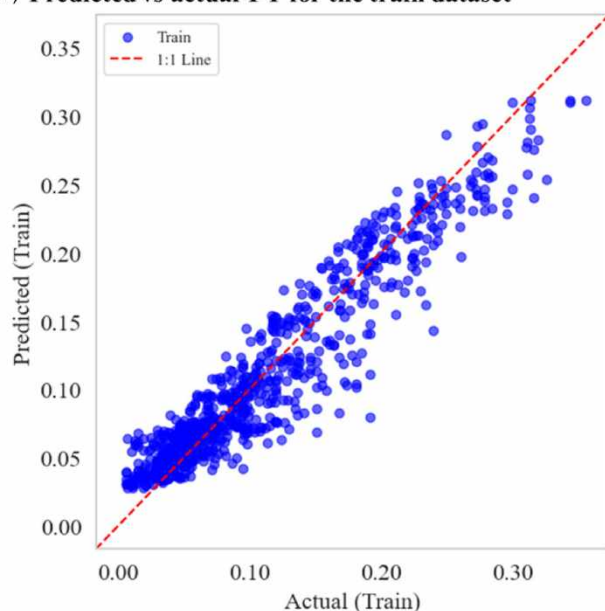
Collectively, the present results and supporting evidence from prior studies indicate that Random Forest is the most reliable, accurate, and robust model for full-scale DAF effluent T-P prediction, making it a suitable choice for data-driven operational decision-making.

[Figure 6](#) illustrates the prediction performance of the optimal Random Forest model through (a) full-period time-series comparisons and (b), (c) actual–predicted scatter plots for the training and test datasets. The time-series plot shows that

(a) Time-series comparison of actual and predicted effluent T-P using the optimal model



(b) Predicted vs actual T-P for the train dataset



(c) Predicted vs actual T-P for the test dataset

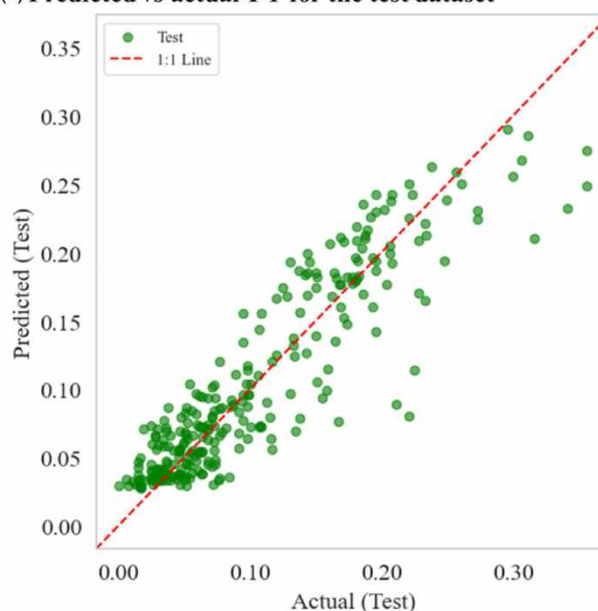


Figure 6 | Prediction performance of the optimal DAF effluent T-P model, including time-series predictions and actual–predicted scatter plots for the train and test datasets.

the model successfully tracks both the baseline trends and short-term fluctuations of effluent T-P concentrations across the entire monitoring period, without exhibiting noticeable lag or systematic bias. Peak values and intermittent shock-type variations typically challenging to predict in daily-scale DAF operation – are also reasonably captured, demonstrating the model's capability to reflect nonlinear and disturbance-driven.

The scatter plots further confirm the strong predictive agreement between observed and estimated values. In the training dataset, predictions are densely aligned along the 1:1 reference line, indicating that the Random Forest model effectively learned the nonlinear mapping between input features and effluent T-P. In the independent test dataset, the predicted values also cluster tightly around the 1:1 line, with no apparent overestimation or underestimation trends, confirming the model's strong generalization capability and robustness against noise.

To further examine whether prediction errors deteriorate under high-loading conditions relevant to chemical dosing support, a regime-based performance evaluation was conducted by stratifying the dataset into effluent T-P concentration quartiles. The low- and high-concentration regimes, defined by the first and third quartiles, comprised 281 and 275 samples, respectively. In the low-concentration regime, the model achieved an RMSE of 0.0214 mg/L and an MAE of 0.0167 mg/L. In contrast, the RMSE and MAE increased to 0.0424 and 0.0328 mg/L, respectively, in the high-concentration regime.

Although RMSE and MAE in the upper quartile were approximately twice those observed in the lower quartile, this increase reflects the inherently higher variability and disturbance intensity associated with high-loading operational

conditions. Importantly, the absolute error magnitudes remain small relative to the observed effluent T-P range, and no evidence of disproportionate error amplification is observed at the upper tail of the concentration distribution. Consistent with the time-series and scatter plot patterns shown in Figure 6, these results indicate that the Random Forest model maintains stable predictive performance even during high-concentration periods where compliance risk is typically concentrated.

From an operational perspective, this regime-based robustness is critical for chemical dosing support, as accurate prediction under high-loading conditions directly influences dosing adjustment and risk mitigation. The combined visual and quantitative evidence therefore confirms that the proposed model is dependable not only under average operating conditions but also during high-variability and high-risk scenarios encountered in full-scale DAF systems.

Overall, these results validate that the Random Forest model provides accurate and stable predictions of daily effluent T-P concentrations, making it suitable for supporting data-driven process monitoring and operational decision-making in full-scale DAF systems.

To quantitatively substantiate the absence of noticeable lag or systematic bias suggested by the time-series and scatter plot analyses, a residual-based diagnostic evaluation was conducted using the independent test dataset, and the results are summarized in Table 6. The regression analysis between predicted and observed effluent T-P concentrations yielded a slope of 0.835, indicating a strong linear relationship between model predictions and observations across the test range. The associated confidence interval confirms that this relationship is statistically stable and does not exhibit substantial deviation from linearity.

The regression intercept was estimated as 0.0175 mg/L, suggesting only a minor offset at low concentration levels. In addition, the mean signed error was calculated as 0.0002 mg/L, demonstrating that the overall prediction bias of the Random Forest model is practically negligible. Together, these residual-based diagnostics indicate that the model does not exhibit systematic overestimation or underestimation tendencies.

3.4. Interpretation of the developed model using explainability methods

Figure 7 presents the SHAP-based feature importance analysis for the optimal Random Forest model, providing quantitative insight into how each input variable contributes to effluent T-P prediction. As shown in Figure 7(a) and 7(b), influent T-P emerged as the most influential factor, exhibiting the highest mean absolute SHAP value (0.009–0.010 mg/L), which is substantially larger than that of any other variable.

Overall, influent T-P emerged as the most influential factor, exhibiting the highest mean absolute SHAP value. This indicates that upstream phosphorus loading acts as the primary driver of daily effluent T-P variability in the DAF process. Coagulant dosage ranked second, with a mean absolute SHAP value of approximately 0.005–0.006 mg/L, confirming that chemical dosing intensity directly governs phosphorus precipitation efficiency and thereby strongly affects the predicted effluent concentration (Wang *et al.* 2021b).

Seasonality, represented by the month variable, showed a moderate but consistent contribution (mean |SHAP| $\approx$ 0.0015–0.002 mg/L), placing it as the third most influential predictor. This reflects the influence of temperature-driven reaction kinetics and seasonal variations in influent water quality. Operational indicators such as DAF inflow, air temperature, influent SS, and the air-to-flow ratio (DAF_A/F) exhibited lower yet non-negligible contributions (mean |SHAP| generally below 0.001 mg/L), suggesting that hydraulic load, climate conditions, and particulate characteristics interact with chemical dosing and influent T-P to shape overall DAF performance. In contrast, IRFR and rainfall showed comparatively minor influence, indicating that their effects on daily effluent T-P are relatively weak or indirect within the studied operating range.

To evaluate the robustness of feature importance across seasons, Figure 7(c) summarizes the seasonal ranking stability of key predictors. Influent T-P consistently ranked as the most important variable in all seasons, while coagulant dosage and month also maintained stable high rankings throughout the year. This seasonal consistency indicates that the dominance of

Table 6 | Residual-based diagnostic results for the Random Forest model using the independent test dataset

Division	Value
Regression slope (95% Confidence Interval) ^a	0.835 (0.789–0.882)
Intercept (95% Confidence Interval) ^a	0.0175 (0.0115–0.0235)
Mean signed error (bias)	0.0002 mg/L

^aConfidence intervals were calculated at the 95% level using the independent test dataset.

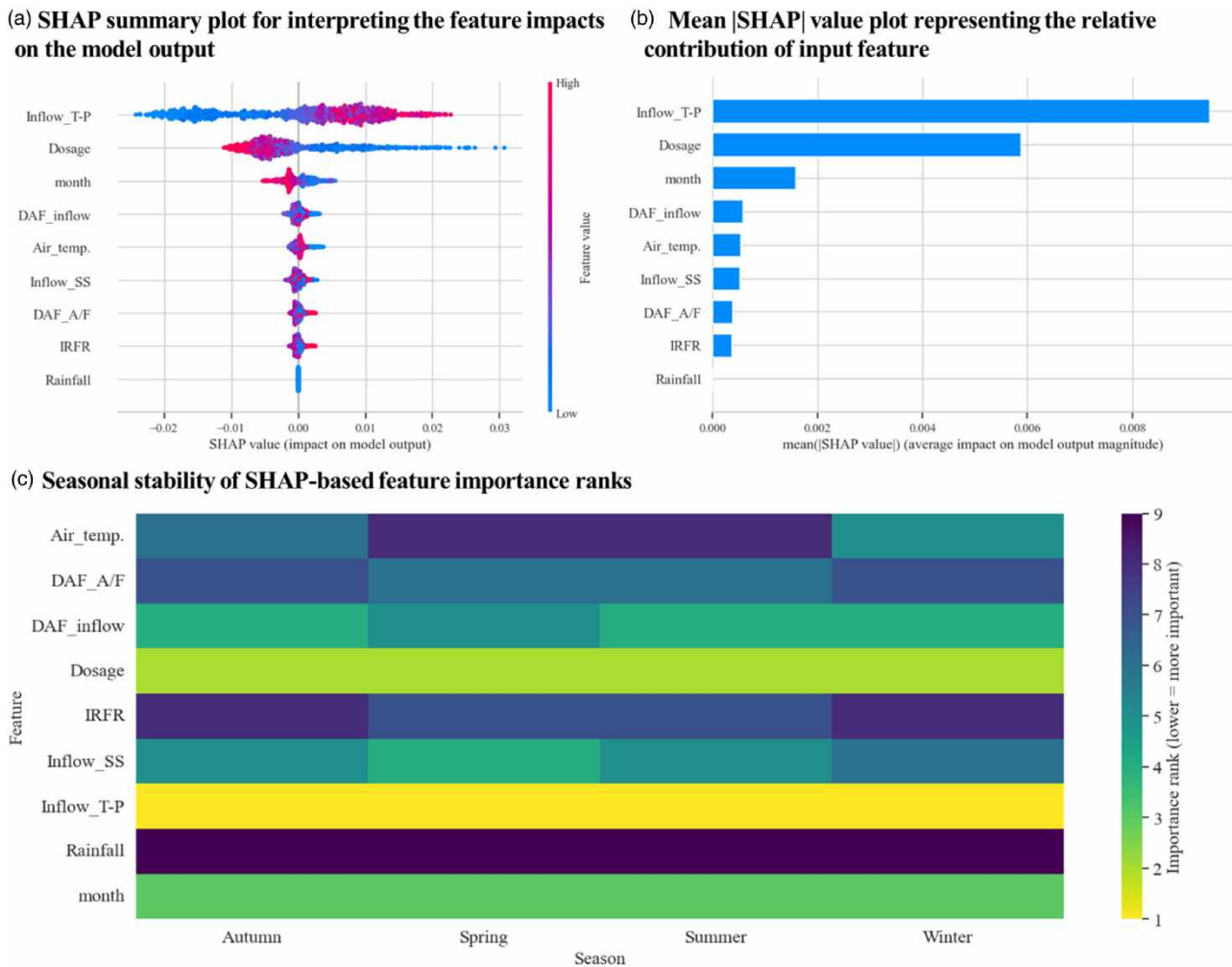


Figure 7 | SHAP-based feature importance analysis results for the optimal DAF effluent T-P prediction model.

influent loading and chemical dosing persists under varying operating conditions rather than being confined to a specific season.

Seasonal and operational indicators are therefore interpreted as reflecting underlying process conditions rather than acting as direct causal drivers. For example, the month and air temperature variables reflect temperature-dependent coagulation kinetics and flotation efficiency, while DAF inflow and DAF_{A/F} capture hydraulic loading and gas–liquid contact conditions that modulate removal performance.

The SHAP summary plot additionally reveals the directionality of feature effects: higher influent T-P and lower coagulant dosage consistently increase predicted effluent T-P concentrations, while seasonal and operational variables exert more complex, nonlinear influences. These directional patterns are consistent with established understanding of DAF phosphorus removal behaviour, indicating that the learned relationships are physically plausible within the operating conditions considered.

Figure 8 presents the SHAP interaction analysis focusing on the three most influential variables identified in the feature-importance results – influent T-P, coagulant dosage, and month. The interaction patterns demonstrate how seasonal effects modulate the contributions of the two dominant predictors, providing deeper insight into the nonlinear behaviour of effluent T-P in the DAF process.

Interaction with influent T-P (a) shows that higher influent phosphorus concentrations consistently lead to positive SHAP contributions across all months, confirming its role as the most critical driver of effluent T-P variability. While this positive

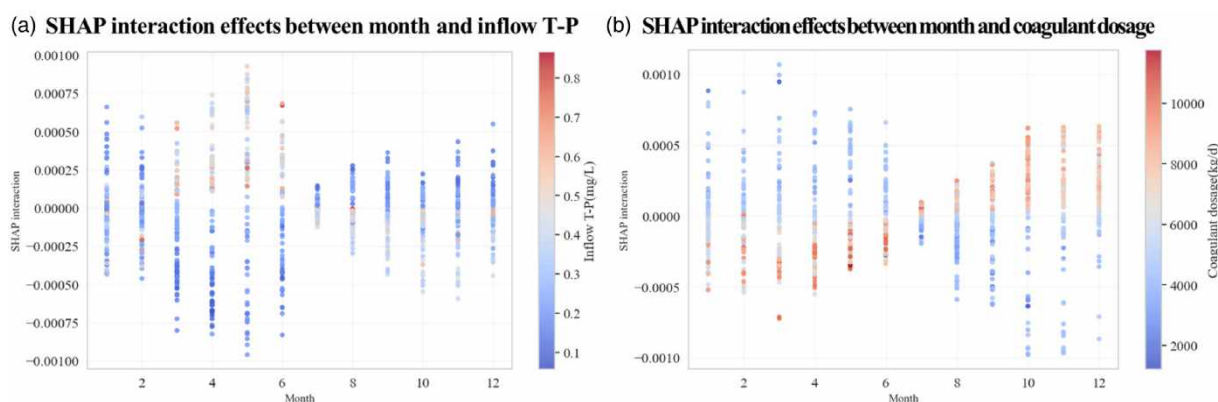


Figure 8 | SHAP-based feature interaction analysis results for the optimal model.

relationship is observed throughout the year, the magnitude of the contribution varies modestly by season, with slightly stronger sensitivity under colder conditions.

In contrast, the interaction between coagulant dosage and month (b) displays a predominantly negative SHAP contribution for higher dosage values, indicating enhanced T-P removal performance as chemical dosing increases. This negative contribution is more pronounced during warmer seasons (May–September), reflecting more favourable coagulation and flotation conditions rather than a direct seasonal effect.

Overall, the interaction analysis confirms that the most important predictors – influent T-P, dosage, and seasonality – do not operate independently but instead exhibit season-dependent nonlinear interactions. These results indicate that seasonal variability modulates, rather than overrides, the dominant effects of influent loading and chemical dosing on DAF phosphorus removal performance.

3.5. Sensitivity-based optimization of coagulant dosing determination

The proposed sensitivity-based optimization framework resulted in substantial reductions in coagulant usage throughout the test period. Based on predictive simulations using the trained model, as summarized in Table 7, the optimized strategy decreased the average daily dosing requirements across all seasons by 32% in spring, 49% in summer, 51% in autumn, and 44% in winter. When extrapolated to an annual scale, these simulation-based reductions correspond to an estimated chemical cost savings of approximately 1.53 billion KRW per year, calculated using the actual coagulant purchasing price applied at the study site (24,840 KRW per 20 kg, equivalent to 1,242 KRW/kg), underscoring the strong economic benefits of data-driven dosing control.

The reported cost savings reflect chemical purchasing costs only and do not include secondary operational expenses such as sludge handling, polymer consumption, or downstream treatment impacts. It should be noted that these optimized dosing levels represent recommended operating targets derived from model-based scenario simulations, rather than guaranteed outcomes from field implementation. These results suggest that the conventional operation relied on excessive or inconsistent chemical addition, whereas the optimized algorithm identified a conditionally optimal dosage range under the assumed model accuracy that is sufficient to sustain treatment performance.

Following this reduction in chemical demand, the seasonal behaviour of effluent T-P concentrations was examined to determine whether treatment stability was preserved or enhanced. As illustrated in Figure 9, optimized dosing produced

Table 7 | Seasonal comparison of original and optimized coagulant dosing amounts

Division	Original Dosage (kg/d)	Optimized Dosage (kg/d)
Spring	254,355	173,331
Summer	206,827	105,859
Autumn	233,273	113,628
Winter	222,780	123,716

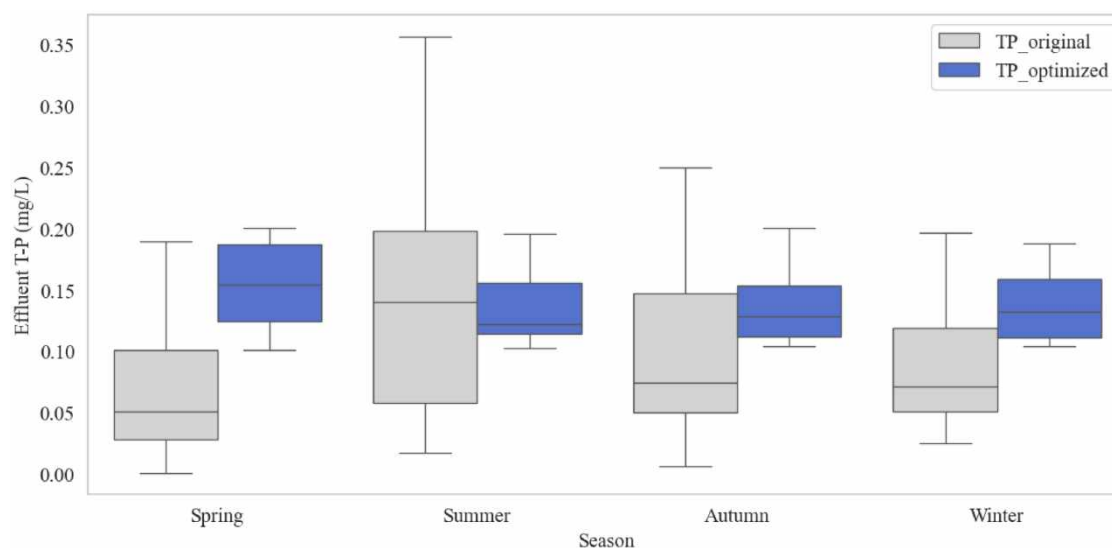


Figure 9 | Seasonal comparison of effluent T-P concentrations under original and optimized coagulant dosing.

markedly narrower seasonal distributions of effluent T-P compared with the original operation, indicating improved process consistency. Across all seasons, the long-tail behaviour and high-end outliers frequently observed under baseline dosing disappeared, and the optimized predictions converged toward the regulatory target range (0.10–0.20 mg/L). The improvement was particularly notable in summer, where high organic and solids loading caused the greatest dispersion under conventional operation; the optimized strategy effectively compressed the distribution and prevented performance deterioration. Likewise, in winter, despite reduced coagulation efficiency caused by low water temperatures the optimized approach maintained a stable and narrow concentration band.

Overall, the sensitivity-based coagulant optimization procedure substantially reduced effluent T-P variability across all seasons while consistently shifting median values toward the target discharge range. This demonstrates that the proposed framework overcomes key limitations of traditional SCADA-based, experience-driven dosing approaches, which often fail to accommodate seasonal loading fluctuations and tend to induce unnecessary over- or under-dosing. The results highlight the strong practical applicability of the proposed method as a robust operational strategy and position it as a core component for developing a data-driven, real-time digital decision-support framework for coagulant dosing optimization.

4. CONCLUSIONS

This study presented an interpretable, feature-engineered machine-learning framework for daily-scale prediction and operational optimization of effluent T-P in a full-scale DAF system. Overall, the results demonstrate that the proposed framework effectively integrates long-term operational data, process-informed feature engineering, and explainable machine learning to provide both reliable prediction and actionable operational insight under realistic wastewater treatment plant conditions.

From a predictive perspective, the Random Forest model exhibited robust and reliable performance, achieving a test R^2 of 0.82 and an RMSE of 0.032 mg/L, corresponding to a prediction error within 20% of the regulatory discharge limit (0.2 mg/L). This level of performance indicates that the model can sufficiently distinguish normal operating conditions from potential compliance-risk scenarios on a daily decision-making horizon, where laboratory-based measurements and routine operational adjustments dominate.

Interpretability analysis using SHAP further revealed that influent T-P concentration and coagulant dosage are the dominant drivers of effluent T-P variability, while short-term variation features and seasonal indicators modulate system response under dynamic loading and environmental conditions. The consistency of these feature contributions across seasons suggests that the learned relationships are physically consistent with established DAF process mechanisms, thereby enhancing model transparency and supporting operator confidence.

Building on these predictive and interpretative results, a sensitivity-based, autoregressive simulation framework was applied to evaluate coagulant dosing strategies. The simulation results indicate that conditionally optimized dosing levels could reduce coagulant consumption by 32–51% across seasons while simultaneously narrowing the effluent T-P distribution toward the target compliance range. The corresponding annual coagulant cost saving was estimated at approximately 1.53 billion KRW, based solely on coagulant purchasing costs. These optimized dosing levels represent model-based recommendations derived from scenario simulations rather than guaranteed outcomes from field implementation, highlighting the role of the framework as a decision-support tool rather than a prescriptive control system.

Taken together, the results indicate that combining daily-scale prediction, explainable machine learning, and sensitivity-based optimization provides a practical and scalable alternative to traditional experience-driven or fixed-ratio dosing strategies. The proposed framework is particularly suited to full-scale facilities where online T-P sensors or fully automated control systems are limited, yet stable regulatory compliance and chemical efficiency are required. Future work will focus on field validation, integration with upstream–downstream process interactions, and extension toward digital-twin and reinforcement-learning-based operational control.

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DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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